```
        set import-rt "1:1"
        set export-rt "1:1"
        set ip-local-learning enable
    next
    edit 84
        set import-rt "84:84"
        set export-rt "84:84"
        set ip-local-learning enable
    next
end
```

2. Configure VXLAN interfaces:

```
config system vxlan
    edit "vx1"
        set interface "loopback1"
        set vni 141
        set evpn-id 1
        set learn-from-traffic enable
    next
     edit "vx84"
        set interface "loopback1"
        set vni 84
        set evpn-id 84
        set learn-from-traffic enable
    next
end
```

3. Configure a system interface:

```
config system interface
    edit "int3-v95"
        set vdom "root"
        set interface "internal3"
        set vlanid 95
    next
end
```

4. Configure system switch interfaces and select the VXLAN interface:

```
config system switch-interface
    edit "sw1"
        set vdom "root"
        set member "int3-v95" "vx1"
        set intra-switch-policy explicit
    next
    edit "sw84"
        set vdom "root"
        set member "vx84"
        set intra-switch-policy explicit
    next
end
```

5. Configure system interfaces:

```
config system interface
    edit "sw1"
        set vdom "root"
        set ip 192.95.1.254 255.255.255.0
        set allowaccess ping ssh telnet
    next
    edit "sw84"
        set vdom "root"
        set ip 192.84.1.254 255.255.255.0
        set allowaccess ping ssh telnet
    next
end
```

6. Configure BGP:

```
config router bgp
    set as 65010
    set router-id 10.255.255.116
    config neighbor
        edit "10.1.2.117"
            set next-hop-self enable
            set soft-reconfiguration-evpn enable
            set remote-as 65010
            set connect-timer 1
        next
    end
    config network
        edit 1
            set prefix 10.255.255.116 255.255.255.255
        next
    end
end
```

To configure FGT117:

1. Configure EVPN:

```
config system evpn
    edit 1
        set import-rt "1:1"
        set export-rt "1:1"
        set ip-local-learning enable
    next
    edit 84
        set import-rt "84:84"
        set export-rt "84:84"
        set ip-local-learning enable
    next
end
```

2. Configure VXLAN interfaces:

```
config system vxlan
    edit "vx1"
        set interface "loopback1"
        set vni 141
        set evpn-id 1
        set learn-from-traffic enable
    next
    edit "vx84"
        set interface "loopback1"
        set vni 84
        set evpn-id 84
        set learn-from-traffic enable
    next
end
```

3. Configure a system switch interface and select the VXLAN interfaces:

```
config system switch-interface
    edit "sw1"
        set vdom "root"
        set member "vx1"
        set intra-switch-policy explicit
    next
    edit "sw84"
        set vdom "root"
        set member "int3-v84"  "vx84"
        set intra-switch-policy explicit
    next
end
```

4. Configure system interfaces:

```
config system interface
    edit "sw1"
        set vdom "root"
        set ip 192.95.1.254 255.255.255.0
        set allowaccess ping ssh telnet
    next
    edit "sw84"
        set vdom "root"
        set ip 192.84.1.254 255.255.255.0
        set allowaccess ping ssh telnet
    next
end
```

5. Configure BGP:

```
config router bgp
    set as 65010
    set router-id 10.255.255.117
    config neighbor
        edit "10.1.2.116"
```

```
            set next-hop-self enable
            set soft-reconfiguration-evpn enable
            set remote-as 65010
            set connect-timer 1
        next
    end
    config network
        edit 1
            set prefix 10.255.255.117 255.255.255.255
        next
    end
end
```

To verify:

1. On FGT116, view the routing table. The FortiGate learned both the subnet of vlan95 and vlan84:

```
# get router info routing-table  all
Codes: K - kernel, C - connected, S - static, R - RIP, B - BGP
       O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       V - BGP VPNv4, E - BGP EVPN
       * - candidate default

Routing table for VRF=0
S*      0.0.0.0/0 [10/0] via 172.18.18.1, wan1, [1/0]
C       10.1.2.0/24 is directly connected, wan2
C       10.255.255.116/32 is directly connected, loopback1
B       10.255.255.117/32 [200/0] via 10.1.2.117 (recursive is directly connected, wan2),
20:28:32, [1/0]
C       172.18.18.0/24 is directly connected, wan1
C       192.84.1.0/24 is directly connected, sw84
C       192.95.1.0/24 is directly connected, sw1
C       192.96.0.0/16 is directly connected, int3-v96
```

2. Look at the L2 EVPN table for VNI 84, FGT116 has learned the MAC address for PC84.
   The MAC address for PC84 is 00:0c:29:da:ca:07 84.

```
# get l2vpn evpn table 84

EVPN instance 84
 Broadcast domain VNI 84 TAGID 0

  EVPN MAC table:
  MAC               VNI      Remote Addr     Binded Address
  00:0c:29:da:ca:07 84       10.255.255.117  -
  00:11:92:d4:5e:8f 84       10.255.255.117  -
  04:d5:90:b1:9d:e8 84       10.255.255.117  -
  6a:a5:45:75:d5:65 84       10.255.255.117  -
```

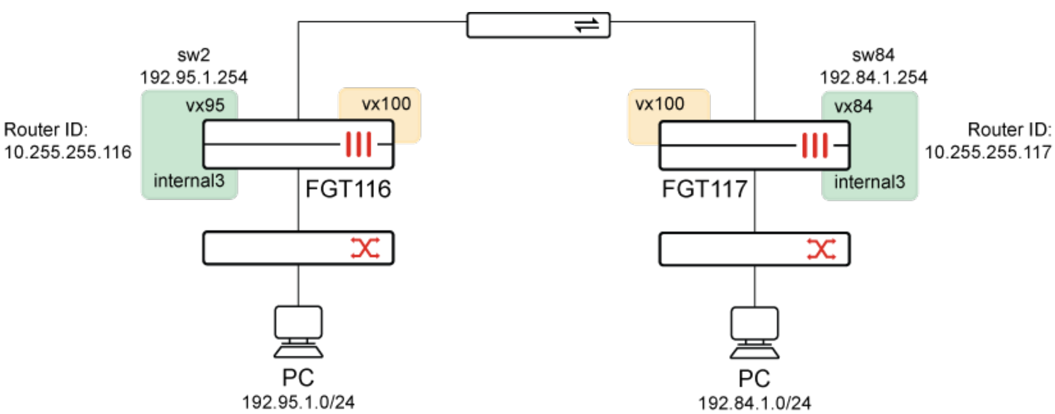
```
EVPN Local MAC table:
"Inactive" means this MAC/IP pair will not be sent to peer.
Flag code: S - Static, F - FDB, L - Switch local MAC, G - Default Gateway.
Trailing * means HA
MAC                Flag Status   Binded Address
02:4c:63:08:3d:39 LG   Inactive 192.84.1.254
                   L    Active   -

EVPN Local IP table:
Address          MAC
192.84.1.254     02:4c:63:08:3d:39

EVPN PEER table:
VNI       Remote Addr      Binded Address
84        10.255.255.117  10.255.255.117
```

Type-5 route example

Instead of advertising MAC/IP bindings used in type-2 routes, type-5 EVPN routes advertise IP prefixes (subnets), allowing EVPN to carry L3 VPN routes across the Security Fabric configured with VXLAN to provide more than only L2 reachability.



This example uses the following VLAN and VNI names and IDs:

PC and L3 instance	VLAN	VNI
PC95	vlan95	vni 95
PC94	vlan84	vni 84
L3 instance	-	vni 100

This example describes how to configure FGT116 and FGT117 in the topology. It also describes how to verify the configuration.

To configure FGT116:

1. Configure EVPN:

```
config system evpn
    edit 95
        set import-rt "95:95"
        set export-rt "95:95"
        set l3-instance 100
    next
    edit 100
        set type ipvrf
        set rd "100:100"
        set import-rt "100:100"
        set export-rt "100:100"
        set interface "loopback1"
    next
end
```

2. Configure VXLAN interfaces:

```
config system vxlan
    edit "vx95"
        set vni 95
        set evpn-id 95
        set learn-from-traffic enable
    next
    edit "vx100"
        set vni 100
        set evpn-id 100
    next
end
```

3. Configure the system switch interface:

```
config system switch-interface
    edit "sw2"
        set vdom "root"
        set member "vx95" "internal3"
        set intra-switch-policy explicit
    next
end
```

4. Configure the system interface:

```
config system interface
    edit "sw2"
        set vdom "root"
        set ip 192.95.1.254 255.255.0.0
        set allowaccess ping
        set type switch
    next
end
```

5. Configure BGP:

```
config router bgp
    set as 65010
    set router-id 10.255.255.116
    config neighbor
        edit "10.1.2.117"
            set next-hop-self enable
            set soft-reconfiguration-evpn enable
            set remote-as 65010
            set connect-timer 1
        next
    end
    config network
        edit 1
            set prefix 10.255.255.116 255.255.255.255
        next
    end
    config redistribute "connected"
        set status evpn enable
    end
end
```

To configure FGT117:

1. Configure EVPN:

```
config system evpn
    edit 100
        set type ipvrf
        set rd "100:100"
        set import-rt "100:100"
        set export-rt "100:100"
        set interface "loopback1"
    next
    edit 84
        set import-rt "84:84"
        set export-rt "84:84"
        set l3-instance 100
    next
end
```

2. Configure VXLAN interfaces:

```
config system vxlan
    edit "vx84"
        set vni 84
        set evpn-id 84
        set learn-from-traffic enable
    next
    edit "vx100"
        set vni 100
```

```
        set evpn-id 100
        set learn-from-traffic enable
    next
end
```

3. Configure the system switch interface:

```
config system switch-interface
    edit "sw84"
        set vdom "root"
        set member "vx84" "internal3"
        set intra-switch-policy explicit
    next
end
```

4. Configure the system interface:

```
config system interface
    edit "sw84"
        set vdom "root"
        set ip 192.84.1.254 255.255.0.0
        set allowaccess ping
        set type switch
    next
end
```

To verify:

1. On FGT117, get router info for the BGP EVPN network:
   Type-5 routes are in bold.

```
FGT117# get router info bgp evpn network
<string>    L2VPN EVPN prefix

FGT117# get router info bgp evpn network
   Network          Next Hop            Metric     LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (Default for VRF 0)
*> [5][0][16][192.84.0.0]/72
                    10.255.255.117      0             100  32768        0 ? <-/1>
*> [5][0][16][192.95.0.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*> [5][0][16][192.96.0.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*  [5][0][16][192.96.0.0]/72
                    10.255.255.117      0             100  32768        0 ? <-/->
*> [5][0][24][10.1.2.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*  [5][0][24][10.1.2.0]/72
                    10.255.255.117      0             100  32768        0 ? <-/->
*> [5][0][24][172.18.18.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*  [5][0][24][172.18.18.0]/72
```

```
                    10.255.255.117      0             100  32768        0 ? <-/->
*> [5][0][32][10.255.255.116]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*> [5][0][32][10.255.255.117]/72
                    10.255.255.117      0             100  32768        0 ? <-/1>

   Network          Next Hop            Metric     LocPrf Weight RouteTag Path
Route Distinguisher: 100:100 (received from VRF 0)
*>i[5][0][16][192.95.0.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*>i[5][0][16][192.96.0.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*>i[5][0][24][10.1.2.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*>i[5][0][24][172.18.18.0]/72
                    10.255.255.116      0             100      0        0 ? <-/1>
*>i[5][0][32][10.255.255.116]/72
                    10.255.255.116      0             100      0        0 ? <-/1>

   Network          Next Hop            Metric     LocPrf Weight RouteTag Path
Route Distinguisher: 0.0.0.0:84 (Default for VRF 0)
*> [2][0][48][00:0c:29:da:ca:07][0]/72
                    10.255.255.117      0             100  32768        0 i <-/1>
*> [2][0][48][00:11:92:d4:5e:8f][0]/72
                    10.255.255.117      0             100  32768        0 i <-/1>
*> [2][0][48][04:d5:90:b1:9d:e8][0]/72
                    10.255.255.117      0             100  32768        0 i <-/1>
*> [2][0][48][0e:61:aa:b2:bb:c3][0]/72
                    10.255.255.117      0             100  32768        0 i <-/1>
*> [3][0][32][10.255.255.117]/80
                    10.255.255.117      0             100  32768        0 i <-/1>
```

2. Get router info for the BGP EVPN network:

```
FGT117# get router info bgp evpn network [5][0][16][192.95.0.0]/72
Route Distinguisher: 100:100 (Default for VRF 0)
BGP routing table entry for [5][0][16][192.95.0.0]/72
  Original VRF 0
  Local
    10.255.255.116 from 10.1.2.116 (10.255.255.116)
      Origin incomplete distance 200 metric 0, localpref 100, valid, internal, evpn, best
      Extended Community: RT:100:100  RMAC:d2:22:98:41:e5:ba
      RD [100:100]
      LABEL1 100
      Last update: Fri Apr 11 14:08:56 2025

Route Distinguisher: 100:100 (received from VRF 0)
BGP routing table entry for [5][0][16][192.95.0.0]/72
  Original VRF 0
  Local
    10.255.255.116 from 10.1.2.116 (10.255.255.116)
      Origin incomplete distance 200 metric 0, localpref 100, valid, internal, evpn, best
```

```
        Extended Community: RT:100:100  RMAC:d2:22:98:41:e5:ba
        RD [100:100]
        LABEL1 100
        Last update: Fri Apr 11 14:08:56 2025
```

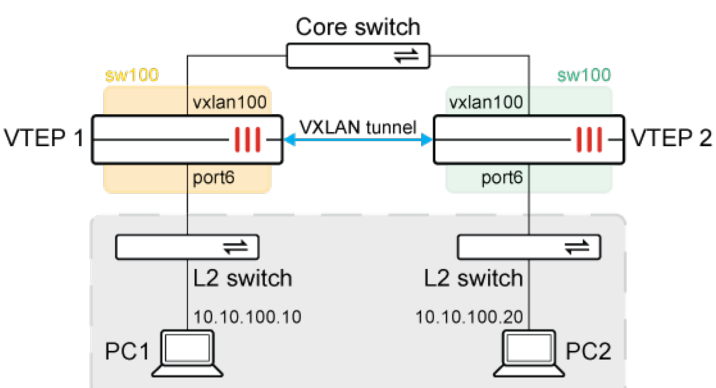
VXLAN troubleshooting

The following commands can be used to troubleshoot VXLAN connectivity:

- `diagnose sys vxlan fdb list <VXLAN_interface>`
- `diagnose sys vxlan fdb stat <VXLAN_interface>`
- `diagnose netlink brctl name host <switch_interface>`
- `diagnose sniffer packet any 'udp and port 4789' 4 0 l`
- `diagnose debug enable`
- `diagnose debug flow filter port 4789`
- `diagnose debug flow trace start <repeat_#>`

Topology

The following topology is used as an example configuration to demonstrate VXLAN troubleshooting steps.



In this example, two FortiGates are configured as VXLAN tunnel endpoints (VTEPs). A VXLAN is configured to allow L2 connectivity between the networks behind each FortiGate. The VXLAN interface and port6 are placed on the same L2 network using a software switch (sw100). An L2 network is formed between PC1 and PC2.

The VTEPs have the following MAC address tables:

Interface/endpoint	VTEP 1	VTEP 2
vxlan100	7e:f2:d1:84:75:0f	ca:fa:31:23:8d:c1

Interface/endpoint	VTEP 1	VTEP 2
port6	00:0c:29:4e:5c:1c	00:0c:29:d0:3e:0d
sw100	00:0c:29:4e:5c:1c	00:0c:29:d0:3e:0d

The MAC address of PC1 is 00:0c:29:90:4f:bf. The MAC address of PC2 is 00:0c:29:f0:88:2c.

To configure the VTEP 1 FortiGate:

1. Configure the local interface:

```
config system vxlan
    edit "vxlan100"
        set interface "port2"
        set vni 100
        set remote-ip "192.168.2.87"
    next
end
```

2. Configure the interface settings:

```
config system interface
    edit "port2"
        set vdom "root"
        set ip 192.168.2.86 255.255.255.0
        set allowaccess ping https ssh http fabric
    next
    edit "vxlan100"
        set vdom "root"
        set type vxlan
        set interface "port2"
    next
end
```

3. Configure the software switch:

```
config system switch-interface
    edit "sw100"
        set vdom "root"
        set member "port6" "vxlan100"
    next
end
```

4. Configure the software switch interface settings:

```
config system interface
    edit "sw100"
        set vdom "root"
        set ip 10.10.100.86 255.255.255.0
        set allowaccess ping
        set type switch
        set device-identification enable
```

```
        set lldp-transmission enable
        set role lan
    next
end
```

To configure the VTEP 2 FortiGate:

1. Configure the local interface:

```
config system vxlan
    edit "vxlan100"
        set interface "port2"
        set vni 100
        set remote-ip "192.168.2.86"
    next
end
```

2. Configure the interface settings:

```
config system interface
    edit "port2"
        set vdom "root"
        set ip 192.168.2.87 255.255.255.0
        set allowaccess ping https ssh snmp http
    next
    edit "vxlan100"
        set vdom "root"
        set type vxlan
        set interface "port2"
    next
end
```

3. Configure the software switch:

```
config system switch-interface
    edit "sw100"
        set vdom "root"
        set member "port6" "vxlan100"
    next
end
```

4. Configure the software switch interface settings:

```
config system interface
    edit "sw100"
        set vdom "root"
        set ip 10.10.100.87 255.255.255.0
        set allowaccess ping
        set type switch
        set device-identification enable
        set lldp-transmission enable
        set role lan
```

```
        set snmp-index 42
    next
end
```

To run diagnostics and debugs:

1. Start a ping from PC1 10.10.100.10 to PC2 10.10.100.20:

```
C:\Users\fortidocs>ping 10.10.100.20

Pinging 10.10.100.20 with 32 bytes of data:
Reply from 10.10.100.20: bytes=32 time=2ms TTL=128
Reply from 10.10.100.20: bytes=32 time=1ms TTL=128
Reply from 10.10.100.20: bytes=32 time=1ms TTL=128
Reply from 10.10.100.20: bytes=32 time<1ms TTL=128

Ping statistics for 10.10.100.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 1ms
```

2. Verify the ARP table:

```
C:\Users\fortidocs>arp /a

Interface: 10.10.100.10 --- 0x21
  Internet Address      Physical Address      Type
  10.10.100.20          00-0c-29-f0-88-2c     dynamic
  10.10.100.86          00-0c-29-4e-5c-1c     dynamic
  10.10.100.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.252           01-00-5e-00-00-fc     static
```

3. Run diagnostics on the VTEP 1 FortiGate.
   a. Verify the forwarding database of VXLAN interface vxlan100:

   ```
   # diagnose sys vxlan fdb list vxlan100
   mac=00:00:00:00:00:00 state=0x0082 remote_ip=192.168.2.87 port=4789 vni=100 ifindex=6
   mac=00:0c:29:f0:88:2c state=0x0002 remote_ip=192.168.2.87 port=4789 vni=100 ifindex=6

   total fdb num: 2
   ```

   The MAC address 00:0c:29:f0:88:2c is learned from PC2 10.10.100.20.
   b. Verify the summary of statistics from the VXLAN's forwarding database:

   ```
   # diagnose sys vxlan fdb stat vxlan100
   fdb_table_size=256 fdb_table_used=2 fdb_entry=2 fdb_max_depth=1 cleanup_idx=0 cleanup_
   timer=252
   ```

   c. Verify the software switch's forwarding table:

```
# diagnose netlink brctl name host sw100
show bridge control interface sw100 host.
fdb: hash size=32768, used=6, num=6, depth=1, gc_time=4, ageing_time=3, simple=switch
Bridge sw100 host table
port no device  devname mac addr                ttl     attributes
  1     7       port6   00:0c:29:4e:5c:1c       0       Local Static
  2     33      vxlan100        7e:f2:d1:84:75:0f       0       Local Static
  2     33      vxlan100        00:00:00:00:00:00       26       Hit(26)
  1     7       port6   00:0c:29:90:4f:bf       0        Hit(0)
  1     7       port6   00:0c:29:d0:3e:ef       7        Hit(7)
  2     33      vxlan100        00:0c:29:f0:88:2c       0        Hit(0)
```

The MAC address of port6 is 00:0c:29:4e:5c:1c. The MAC address of vxlan100 is 7e:f2:d1:84:75:0f. The MAC address 00:0c:29:f0:88:2c of PC2 is learned from the remote network.

4. Run diagnostics on the VTEP 2 FortiGate.
   a. Verify the forwarding database of VXLAN interface vxlan100:

```
# diagnose sys vxlan fdb list vxlan100
mac=00:00:00:00:00:00 state=0x0082 remote_ip=192.168.2.86 port=4789 vni=100 ifindex=6
mac=00:0c:29:90:4f:bf state=0x0002 remote_ip=192.168.2.86 port=4789 vni=100 ifindex=6

total fdb num: 2
```

   The MAC address 00:0c:29:90:4f:bf is learned from PC1 10.10.100.10.

   b. Verify the summary of statistics from the VXLAN's forwarding database:

```
# diagnose sys vxlan fdb stat vxlan100
fdb_table_size=256 fdb_table_used=2 fdb_entry=2 fdb_max_depth=1 cleanup_idx=0 cleanup_
timer=304
```

   c. Verify the software switch's forwarding table:

```
# diagnose netlink brctl name host sw100
show bridge control interface sw100 host.
fdb: hash size=32768, used=5, num=5, depth=1, gc_time=4, ageing_time=3, simple=switch
Bridge sw100 host table
port no device  devname mac addr                ttl     attributes
  2     50      vxlan100        00:00:00:00:00:00       10       Hit(10)
  2     50      vxlan100        00:0c:29:90:4f:bf       2        Hit(2)
  1     7       port6   00:0c:29:d0:3e:0d       0       Local Static
  2     50      vxlan100        ca:fa:31:23:8d:c1       0       Local Static
  1     7       port6   00:0c:29:f0:88:2c       0        Hit(0)
```

   The MAC address of port6 is 00:0c:29:d0:3e:0d. The MAC address of vxlan100 is ca:fa:31:23:8d:c1. The MAC address 00:0c:29:90:4f:bf of PC1 is learned from the remote network.

5. Perform a sniffer trace on the VTEP 1 FortiGate to view the life of the packets as they pass through the FortiGate:

```
# diagnose sniffer packet any 'host 10.10.100.20 or (udp and host 192.168.2.87)' 4 0 l
Using Original Sniffing Mode
interfaces=[any]
```

```
filters=[host 10.10.100.20 or (udp and host 192.168.2.87)]
2022-11-04 14:35:18.567602 port6 in arp who-has 10.10.100.20 tell 10.10.100.10
2022-11-04 14:35:18.567629 vxlan100 out arp who-has 10.10.100.20 tell 10.10.100.10
2022-11-04 14:35:18.567642 port2 out 192.168.2.86.4804 -> 192.168.2.87.4789: udp 68
2022-11-04 14:35:18.567658 sw100 in arp who-has 10.10.100.20 tell 10.10.100.10
2022-11-04 14:35:18.568239 port2 in 192.168.2.87.4789 -> 192.168.2.86.4789: udp 68
2022-11-04 14:35:18.568263 vxlan100 in arp reply 10.10.100.20 is-at 00:0c:29:f0:88:2c
2022-11-04 14:35:18.568272 port6 out arp reply 10.10.100.20 is-at 00:0c:29:f0:88:2c
2022-11-04 14:35:18.568425 port6 in 10.10.100.10 -> 10.10.100.20: icmp: echo request
2022-11-04 14:35:18.568435 vxlan100 out 10.10.100.10 -> 10.10.100.20: icmp: echo request
2022-11-04 14:35:18.568443 port2 out 192.168.2.86.4805 -> 192.168.2.87.4789: udp 82
2022-11-04 14:35:18.568912 port2 in 192.168.2.87.4789 -> 192.168.2.86.4789: udp 68
2022-11-04 14:35:18.568925 vxlan100 in arp who-has 10.10.100.10 tell 10.10.100.20
2022-11-04 14:35:18.568935 port6 out arp who-has 10.10.100.10 tell 10.10.100.20
2022-11-04 14:35:18.568945 sw100 in arp who-has 10.10.100.10 tell 10.10.100.20
2022-11-04 14:35:18.569070 port6 in arp reply 10.10.100.10 is-at 00:0c:29:90:4f:bf
2022-11-04 14:35:18.569076 vxlan100 out arp reply 10.10.100.10 is-at 00:0c:29:90:4f:bf
2022-11-04 14:35:18.569081 port2 out 192.168.2.86.4806 -> 192.168.2.87.4789: udp 68
2022-11-04 14:35:18.569417 port2 in 192.168.2.87.4789 -> 192.168.2.86.4789: udp 82
2022-11-04 14:35:18.569427 vxlan100 in 10.10.100.20 -> 10.10.100.10: icmp: echo reply
2022-11-04 14:35:18.569431 port6 out 10.10.100.20 -> 10.10.100.10: icmp: echo reply
```

In the output, the following packet sequence is seen on the FortiGate:

a. The FortiGate receives an ARP request from PC1 10.10.100.10 on port6.
b. The ARP request is forwarded to vxlan100 on the same software switch, where it gets encapsulated and sent out as a UDP port 4789 packet on port2.
c. A reply is received on port2 from the remote VTEP with the ARP response encapsulated in UDP port 4789 again.
d. The ARP reply is forwarded back out of port6 to PC1.
e. PC1 sends the ICMP request using the same steps.

6. Perform the same sniffer trace filter with a level 6 verbose level. In this example, the packet capture is converted into a Wireshark file.